# THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679


## ANSWER KEY & MARKING SCHEME SET 1

Chapter 2: **Arithmetic Expressions** | Class 7 Maths | Maximum Marks: 40

### SECTION — A (Objective Type) — Answers — 10 × 1 = 10

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(c)** 36 | Apply BODMAS: brackets, orders, division/multiplication, addition/subtraction. |
| 2 | **(d)** 27 | Apply BODMAS: brackets, orders, division/multiplication, addition/subtraction. |
| 3 | **(c)** 50 | Evaluate first. |
| 4 | **(a)** 9 | Divide the product by the multiplier, then subtract. |
| 5 | **(c)** 325 | Total minus reserved. |
| 6 | **(d)** $20 - 5 \times 2$ | Evaluate each option. |
| 7 | **(c)** 15 | Apply BODMAS: brackets, orders, division/multiplication, addition/subtraction. |
| 8 | **(a)** 28 | Apply BODMAS: brackets, orders, division/multiplication, addition/subtraction. |
| 9 | **(b)** 69 | Apply BODMAS: brackets, orders, division/multiplication, addition/subtraction. |
| 10 | **(d)** 22 | Evaluate first. |

### SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 11 | Innermost bracket first; Then the next bracket (1); Then the rest in order; Value $= 18$ (1). | 1+1 |
| 12 | $6 \times (9 + 8)$ (1); Value $= 102$ (1). | 1+1 |
| 13 | First $= 60$; Second $= 60$ (1); Equal (1). | 1+1 |
| 14 | Innermost bracket first; Then the next bracket (1); Then the rest in order; Value $= 46$ (1). | 1+1 |

### SECTION — C (Short Answer) — Answers — 4 × 3 = 12

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 15 | Cost $= 50 \times 40 = 2000$; Sales $= 30 \times 60 + (50 - 30) \times 35 = 2500$ (1); Result $= 500$ (1); Profit (1). | 1+1+1 |
| 16 | Brackets first; Multiplication (1); Final answer (1); Value $= 180$ (1). | 1+1+1 |
| 17 | Try bracketing different parts (1); Correct: $(2 + 3) \times 4 + 5$ (1); Check value 25 (1). | 1+1+1 |
| | **OR:** Cost $= 40 \times 55 = 2200$; Sales $= 25 \times 70 + (40 - 25) \times 45 = 2425$ (1); Result $= 225$ (1); Profit (1). | 1+1+1 |
| 18 | Brackets first; Multiplication (1); Final answer (1); Value $= 120$ (1). | 1+1+1 |
| | **OR:** Try bracketing different parts (1); Correct: $(10 - 4) \div 2 + 3$ (1); Check value 6 (1). | 1+1+1 |

### SECTION — D (Long Answer) — Answers — 2 × 5 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 19 | Chairs after discount $= 40 \times (450 - 50) = 16000$ (1); Tables $= 10 \times 1200 = 12000$ (1); Total $= 28000$ (1); Expression: $40 \times (450 - 50) + 10 \times 1200$ (2). | 1+1+1+2 |
| | **OR:** Evaluate brackets first (1); Then multiplication/division (1); Value $= 465$ (1); A suitable story is consistent with the operations (2). | 1+1+1+2 |
| 20 | Chairs after discount $= 30 \times (600 - 100) = 15000$ (1); Tables $= 8 \times 1500 = 12000$ (1); Total $= 27000$ (1); Expression: $30 \times (600 - 100) + 8 \times 1500$ (2). | 1+1+1+2 |
| | **OR:** Evaluate brackets first (1); Then multiplication/division (1); Value $= 80$ (1); A suitable story is consistent with the operations (2). | 1+1+1+2 |

**Mistakes to avoid:**

- Working strictly left to right, e.g. $5 + 3 \times 4 = 32$; multiplication comes first.
- Doing $12 \div 3 \times 2$ as $12 \div 6$; division and multiplication are done left to right.
- Treating subtraction as associative: $20 - (8 - 4) \neq 20 - 8 - 4$.
- Using the distributive property wrongly: $a \times (b + c) \neq a \times b + c$.
- Forgetting to evaluate the bracket first, e.g. taking $2 \times (9 - 4)$ as $18 - 4$.

**Evaluator's note:** Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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## ANSWER KEY & MARKING SCHEME SET 2

Chapter 2: **Arithmetic Expressions** | Class 7 Maths | Maximum Marks: 40

### SECTION — A (Objective Type) — Answers — 10 × 1 = 10

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(a)** 8 | Divide the product by the multiplier, then subtract. |
| 2 | **(b)** 312 | Total minus reserved. |
| 3 | **(a)** $(24 + 8) \div 4$ | Evaluate each option. |
| 4 | **(a)** 60 | Apply BODMAS: brackets, orders, division/multiplication, addition/subtraction. |
| 5 | **(c)** 32 | Apply BODMAS: brackets, orders, division/multiplication, addition/subtraction. |
| 6 | **(c)** 35 | Apply BODMAS: brackets, orders, division/multiplication, addition/subtraction. |
| 7 | **(a)** 41 | Evaluate first. |
| 8 | **(a)** 11 | Divide the product by the multiplier, then subtract. |
| 9 | **(c)** 300 | Total minus reserved. |
| 10 | **(b)** $8 \times (7 - 3)$ | Evaluate each option. |

### SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 11 | $48 \div 6 + 7 \times 5$ (1); Value $= 43$ (1). | 1+1 |
| 12 | First $= 10$; Second $= 10$ (1); Equal (1). | 1+1 |
| 13 | Innermost bracket first; Then the next bracket (1); Then the rest in order; Value $= 8$ (1). | 1+1 |
| 14 | $100 - 8 \times 11$ (1); Value $= 12$ (1). | 1+1 |

### SECTION — C (Short Answer) — Answers — 4 × 3 = 12

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 15 | Cost $= 60 \times 30 = 1800$; Sales $= 35 \times 50 + (60 - 35) \times 25 = 2375$ (1); Result $= 575$ (1); Profit (1). | 1+1+1 |
| 16 | Brackets first; Multiplication (1); Final answer (1); Value $= 144$ (1). | 1+1+1 |
| 17 | Try bracketing different parts (1); Correct: $(6 + 2) \times 5 - 3$ (1); Check value 37 (1). | 1+1+1 |
| | **OR:** Cost $= 45 \times 65 = 2925$; Sales $= 20 \times 90 + (45 - 20) \times 50 = 3050$ (1); Result $= 125$ (1); Profit (1). | 1+1+1 |
| 18 | Brackets first; Multiplication (1); Final answer (1); Value $= 250$ (1). | 1+1+1 |
| | **OR:** Try bracketing different parts (1); Correct: $(8 - 3) \times 2 + 4$ (1); Check value 14 (1). | 1+1+1 |

### SECTION — D (Long Answer) — Answers — 2 × 5 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 19 | Chairs after discount $= 25 \times (520 - 20) = 12500$ (1); Tables $= 12 \times 900 = 10800$ (1); Total $= 23300$ (1); Expression: $25 \times (520 - 20) + 12 \times 900$ (2). | 1+1+1+2 |
| | **OR:** Evaluate brackets first (1); Then multiplication/division (1); Value $= 220$ (1); A suitable story is consistent with the operations (2). | 1+1+1+2 |
| 20 | Chairs after discount $= 50 \times (380 - 30) = 17500$ (1); Tables $= 6 \times 1800 = 10800$ (1); Total $= 28300$ (1); Expression: $50 \times (380 - 30) + 6 \times 1800$ (2). | 1+1+1+2 |
| | **OR:** Evaluate brackets first (1); Then multiplication/division (1); Value $= 50$ (1); A suitable story is consistent with the operations (2). | 1+1+1+2 |

**Mistakes to avoid:**

- Working strictly left to right, e.g. $5 + 3 \times 4 = 32$; multiplication comes first.
- Doing $12 \div 3 \times 2$ as $12 \div 6$; division and multiplication are done left to right.
- Treating subtraction as associative: $20 - (8 - 4) \neq 20 - 8 - 4$.
- Using the distributive property wrongly: $a \times (b + c) \neq a \times b + c$.
- Forgetting to evaluate the bracket first, e.g. taking $2 \times (9 - 4)$ as $18 - 4$.

**Evaluator's note:** Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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## ANSWER KEY & MARKING SCHEME SET 3

Chapter 2: **Arithmetic Expressions** | Class 7 Maths | Maximum Marks: 40

### SECTION — A (Objective Type) — Answers — 10 × 1 = 10

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(a)** 27 | Apply BODMAS: brackets, orders, division/multiplication, addition/subtraction. |
| 2 | **(d)** 29 | Apply BODMAS: brackets, orders, division/multiplication, addition/subtraction. |
| 3 | **(b)** 100 | Apply BODMAS: brackets, orders, division/multiplication, addition/subtraction. |
| 4 | **(d)** 91 | Evaluate first. |
| 5 | **(a)** 7 | Divide the product by the multiplier, then subtract. |
| 6 | **(d)** 325 | Total minus reserved. |
| 7 | **(d)** $48 \div (6 + 2)$ | Evaluate each option. |
| 8 | **(a)** 41 | Apply BODMAS: brackets, orders, division/multiplication, addition/subtraction. |
| 9 | **(b)** 12 | Apply BODMAS: brackets, orders, division/multiplication, addition/subtraction. |
| 10 | **(c)** 25 | Apply BODMAS: brackets, orders, division/multiplication, addition/subtraction. |

### SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 11 | First $= 29$; Second $= 41$ (1); Not equal (1). | 1+1 |
| 12 | Innermost bracket first; Then the next bracket (1); Then the rest in order; Value $= 60$ (1). | 1+1 |
| 13 | $2 \times (15 + 20) - 30$ (1); Value $= 40$ (1). | 1+1 |
| 14 | First $= 10$; Second $= 40$ (1); Not equal (1). | 1+1 |

### SECTION — C (Short Answer) — Answers — 4 × 3 = 12

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 15 | Cost $= 80 \times 25 = 2000$; Sales $= 48 \times 40 + (80 - 48) \times 20 = 2560$ (1); Result $= 560$ (1); Profit (1). | 1+1+1 |
| 16 | Brackets first; Multiplication (1); Final answer (1); Value $= 180$ (1). | 1+1+1 |
| 17 | Try bracketing different parts (1); Correct: $(4 + 5) \times 2 - 6$ (1); Check value 12 (1). | 1+1+1 |
| | **OR:** Cost $= 36 \times 48 = 1728$; Sales $= 24 \times 70 + (36 - 24) \times 38 = 2136$ (1); Result $= 408$ (1); Profit (1). | 1+1+1 |
| 18 | Brackets first; Multiplication (1); Final answer (1); Value $= 70$ (1). | 1+1+1 |
| | **OR:** Try bracketing different parts (1); Correct: $(9 - 3) \times 4 + 2$ (1); Check value 26 (1). | 1+1+1 |

### SECTION — D (Long Answer) — Answers — 2 × 5 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 19 | Chairs after discount $= 20 \times (700 - 150) = 11000$ (1); Tables $= 5 \times 2500 = 12500$ (1); Total $= 23500$ (1); Expression: $20 \times (700 - 150) + 5 \times 2500$ (2). | 1+1+1+2 |
| | **OR:** Evaluate brackets first (1); Then multiplication/division (1); Value $= 315$ (1); A suitable story is consistent with the operations (2). | 1+1+1+2 |
| 20 | Chairs after discount $= 35 \times (480 - 40) = 15400$ (1); Tables $= 9 \times 1100 = 9900$ (1); Total $= 25300$ (1); Expression: $35 \times (480 - 40) + 9 \times 1100$ (2). | 1+1+1+2 |
| | **OR:** Evaluate brackets first (1); Then multiplication/division (1); Value $= 30$ (1); A suitable story is consistent with the operations (2). | 1+1+1+2 |

**Mistakes to avoid:**

- Working strictly left to right, e.g. $5 + 3 \times 4 = 32$; multiplication comes first.
- Doing $12 \div 3 \times 2$ as $12 \div 6$; division and multiplication are done left to right.
- Treating subtraction as associative: $20 - (8 - 4) \neq 20 - 8 - 4$.
- Using the distributive property wrongly: $a \times (b + c) \neq a \times b + c$.
- Forgetting to evaluate the bracket first, e.g. taking $2 \times (9 - 4)$ as $18 - 4$.

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